For $r \neq 0$ the equation in the cylindric coordinates is

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{d\varphi}{dr}\right) - \frac{\varphi}{a^2} = 0$$

$$\varphi'' + \frac{1}{r}\varphi' - \frac{\varphi}{a^2} = 0 \qquad (1)$$

This is a particular example of a more general Bessel equation

$$f'' + \frac{1}{\xi}f' + \left(1 - \frac{n^2}{\xi^2}\right)f = 0 \qquad (2)$$

Eq. (2) reduces to Eq. (1) for $n=0$ and $\xi = ir/a$.

The most convenient solutions of the Bessel equation of imaginary argument are modified Bessel funtions I_n and K_n, which are <u>real</u>. The second function, K_n, is also know as Macdonald function:

$$\varphi(r) = A\, I_0(r/a) + B\, K_0(r/a)$$

The function $I_0(ra)$ exponentially grows when $r \to \infty$. Thus, $A = 0$

The remaining constant B can be found from analyzing the $r \to 0$ behavior:

$$K_0(ra \to 0) = -\ln(r/a)$$

This gives for electric field at small distances

$$E(r \to 0) = -\varphi'(r) = \frac{B}{r}$$

On the other hand the field at $r \to 0$ is <u>unscreened</u> and is given by the fiel of a charged wire:

$$E(r) = \frac{\lambda_f}{2\pi\varepsilon_0 r} \qquad (3)$$

This fixes $B = \frac{\lambda_f}{2\pi\varepsilon_0}$ and gives the solution at arbitrary distances

$$\varphi(r) = \frac{\lambda_f}{2\pi\varepsilon_0} K_0(r/a) \qquad (4)$$

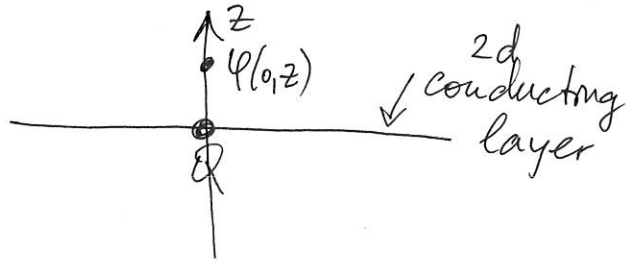
The behavior at $ra \gg 1$ is given by the asymptotics of the function K_0:

$$K_0(\xi \gg 1) \approx \sqrt{\frac{\pi}{2\xi}}\, e^{-\xi}$$

which gives

$$\varphi(r \to \infty) \approx \frac{\lambda_f}{2\varepsilon_0} \frac{e^{-r/a}}{\sqrt{2\pi r/a}}$$

P2] In Lect. 18 we found $\varphi(\vec{q},z) = \dfrac{Q\, e^{-q|z|}}{2\varepsilon_0 (q+\varkappa_2)}$



$$\varphi(\vec{\rho},z) = \int \frac{d^2\vec{q}}{(2\pi)^2}\, \varphi(\vec{q},z)\, e^{i\vec{q}\cdot\vec{\rho}} = \frac{Q}{4\pi\varepsilon_0}\int_0^\infty dq\, \frac{q\, e^{-q|z|}}{q+\varkappa_2}\,\cdot$$

$$\cdot \int_0^{2\pi} \frac{d\theta}{2\pi}\, e^{iq\rho\cos\theta} = \left(\text{use } \frac{1}{q+\varkappa_2} = \int_0^\infty du\, e^{-u(q+\varkappa_2)}\right)$$

$$= +\frac{Q}{4\pi\varepsilon_0}\int_0^\infty du\, e^{-\varkappa_2 u}\int_0^{2\pi}\frac{d\theta}{2\pi}\underbrace{\int_0^\infty dq\, q\, e^{-q(u-i\rho\cos\theta+|z|)}}_{-\frac{\partial}{\partial u}\int_0^\infty dq\, e^{-q(u+|z|-i\rho\cos\theta)}}$$

$$= -\frac{Q}{4\pi\varepsilon_0}\int_0^\infty du\, e^{-\varkappa_2 u}\,\frac{\partial}{\partial u}\int_0^{2\pi}\frac{d\theta}{2\pi}\,\frac{1}{u+|z|-i\rho\cos\theta}$$

Now, at $\vec{\rho}=(0,0)$ everything becomes simple

$$\varphi(0,z) = -\frac{Q}{4\pi\varepsilon_0}\int_0^\infty du\, e^{-\varkappa_2 u}\,\frac{\partial}{\partial u}\,\frac{1}{u+|z|}$$

$$= \frac{Q}{4\pi\varepsilon_0}\int_0^\infty du\, \frac{e^{-\varkappa_2 u}}{(u+|z|)^2} = \left(u = |z|\, x\right)$$

$$= \frac{Q}{4\pi\varepsilon_0 |z|}\int_0^\infty dx\, \frac{e^{-\varkappa_2|z|x}}{(1+x)^2} \Rightarrow \text{for } \varkappa_2|z| \gg 1$$

$$\varphi(0,z) \approx \frac{Q}{4\pi\varepsilon_0|z|}\int_0^\infty dx\, e^{-\varkappa_2|z|x} = \frac{Q}{4\pi\varepsilon_0}\,\frac{1}{\varkappa_2 z^2} \sim \frac{1}{z^2}.$$

while $\varphi(\rho,0) = \frac{Q}{4\pi\varepsilon_0}\,\frac{1}{\rho(\varkappa_2\rho)^2} \sim \frac{1}{\rho^3}$.

Problem 3

$$E_x = A e^{ikx-i\omega t} \begin{cases} \exp\left(-z\sqrt{k^2-\omega^2/c^2}\right), & z>0 \\ \exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right), & z<0 \end{cases} \qquad (1)$$

Similarly, E_z is given in the class notes:

$$E_z = ikAe^{ikx-i\omega t} \begin{cases} \dfrac{\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right)}{\sqrt{k^2-\omega^2/c^2}}, & z>0 \\ -\dfrac{\exp\left(z\sqrt{k^2-\varepsilon\omega^2/\varepsilon_0 c^2}\right)}{\sqrt{k^2-\varepsilon\omega^2/\varepsilon_0 c^2}}, & z<0 \end{cases} \qquad (2)$$

Magnetic field is given by

$$\vec{B} = -\frac{i}{\omega}\nabla\times\vec{E}$$

$$B_x = -\frac{i}{\omega}\left(\frac{\partial}{\partial y}E_z - \frac{\partial E_y}{\partial z}\right) = 0$$

(since $E_y=0$ and E_x, E_z are independent of y)

$$B_z = -\frac{i}{\omega}\left(\frac{\partial}{\partial x}E_y - \frac{\partial E_x}{\partial y}\right) = 0$$

$$B_y = -\frac{i}{\omega}\left(\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}\right) = -\frac{i}{\omega}\left(\frac{\partial E_x}{\partial z} - ikE_z\right) \qquad (3)$$

We now substitute Eqs. (1) and (2) into Eq. (3):

for $z>0$:

$$B_y = -\frac{i}{\omega}\left(-\sqrt{k^2-\omega^2/c^2}\,Ae^{ikx-i\omega t}\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right) + k^2Ae^{ikx-i\omega t}\frac{\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right)}{\sqrt{k^2-\omega^2/c^2}}\right) = \qquad (4)$$

$$= -\frac{i\omega}{c^2} A e^{ikx-i\omega t} \frac{\exp\left(-z\sqrt{k^2-\omega^2/c^2}\right)}{\sqrt{k^2-\omega^2/c^2}}$$

for $z<0$:

$$B_y = -\frac{i}{\omega}\left(\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\, A e^{ikx-i\omega t} \exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right)\right.$$

$$\left. - k^2 A e^{ikx-i\omega t} \frac{\exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right)}{\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}}\right) \qquad (5)$$

$$= \frac{i\varepsilon\omega}{\varepsilon_0 c^2} A e^{ikx-i\omega t} \frac{\exp\left(z\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}\right)}{\sqrt{k^2-\frac{\varepsilon\omega^2}{\varepsilon_0 c^2}}}$$

By virtue of the dispersion equation for the surface plasmons:

$$\frac{\varepsilon}{\varepsilon_0} = -\frac{\sqrt{k^2-\frac{\omega^2\varepsilon}{c^2\varepsilon_0}}}{\sqrt{k^2-\frac{\omega^2}{c^2}}}$$

the two expressions (4) and (5) match at $z=0$.

Thus, $\vec{B}$ is indeed continuous at the surface.